

Evaluating Data Augmentation Under Limited Biological Imaging

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Motivation

- Biological imaging datasets are expensive and time consuming to obtain, limiting the available training data for machine learning models.
- Data augmentation combats these scarcity problems by applying data transformations such as flips, rotations, and erasing to expand the diversity and size of datasets, improving model generalization under limited data constraints.
- However, we hypothesize that whether these techniques meaningfully improve model generalization may depend on the structural properties of the underlying data.
- We evaluate several geometric augmentations on two biomedical datasets — cellular microscopy and lung X-rays — under limited data constraints.

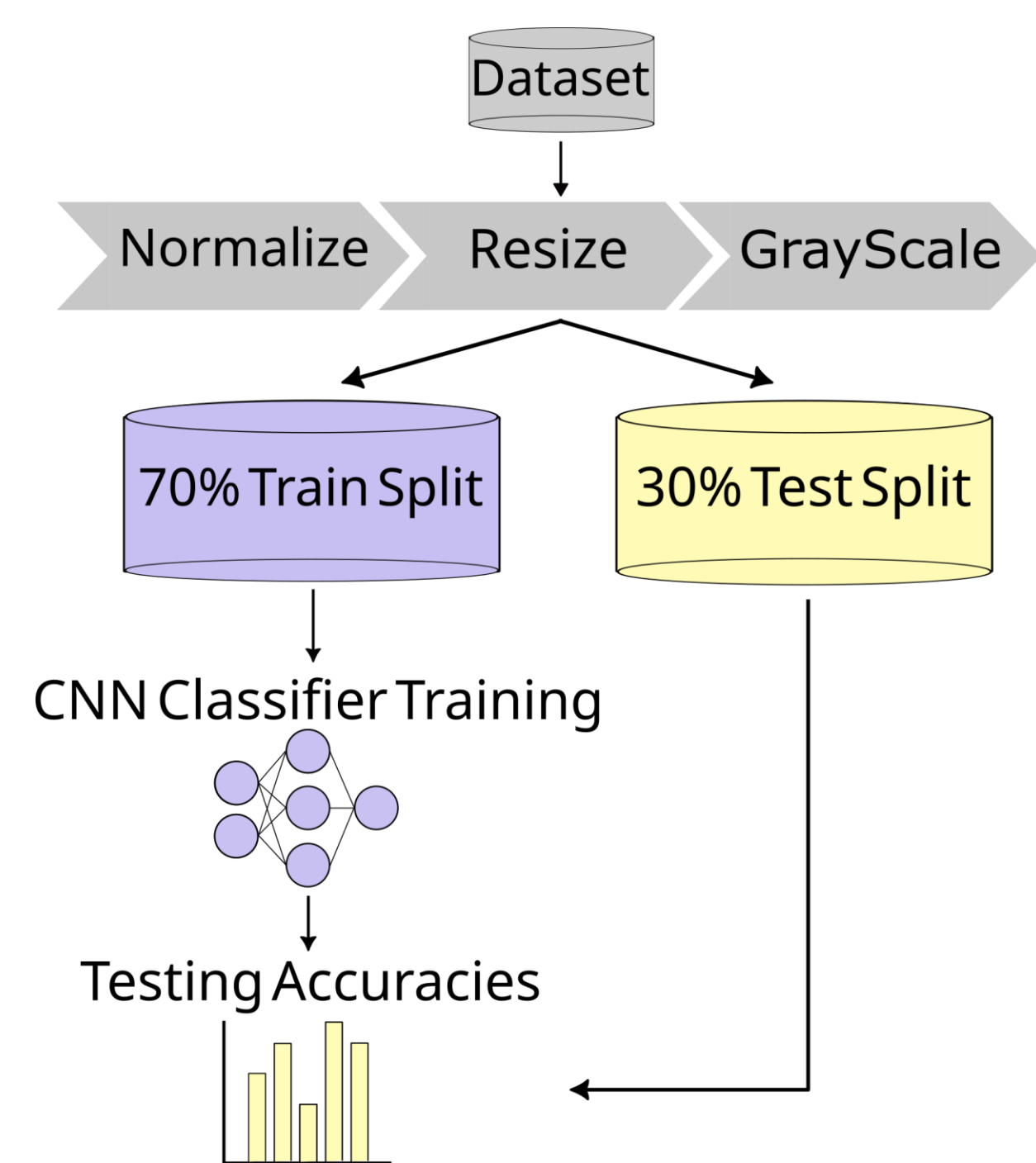


Figure 1. Overview of data preparation and model training pipeline. Data preprocessing included normalizing the images, resizing them to a standard 128x128 pixels, and converting them into single-channel grayscale. The data was then split into a training and testing set before being passed through the model.

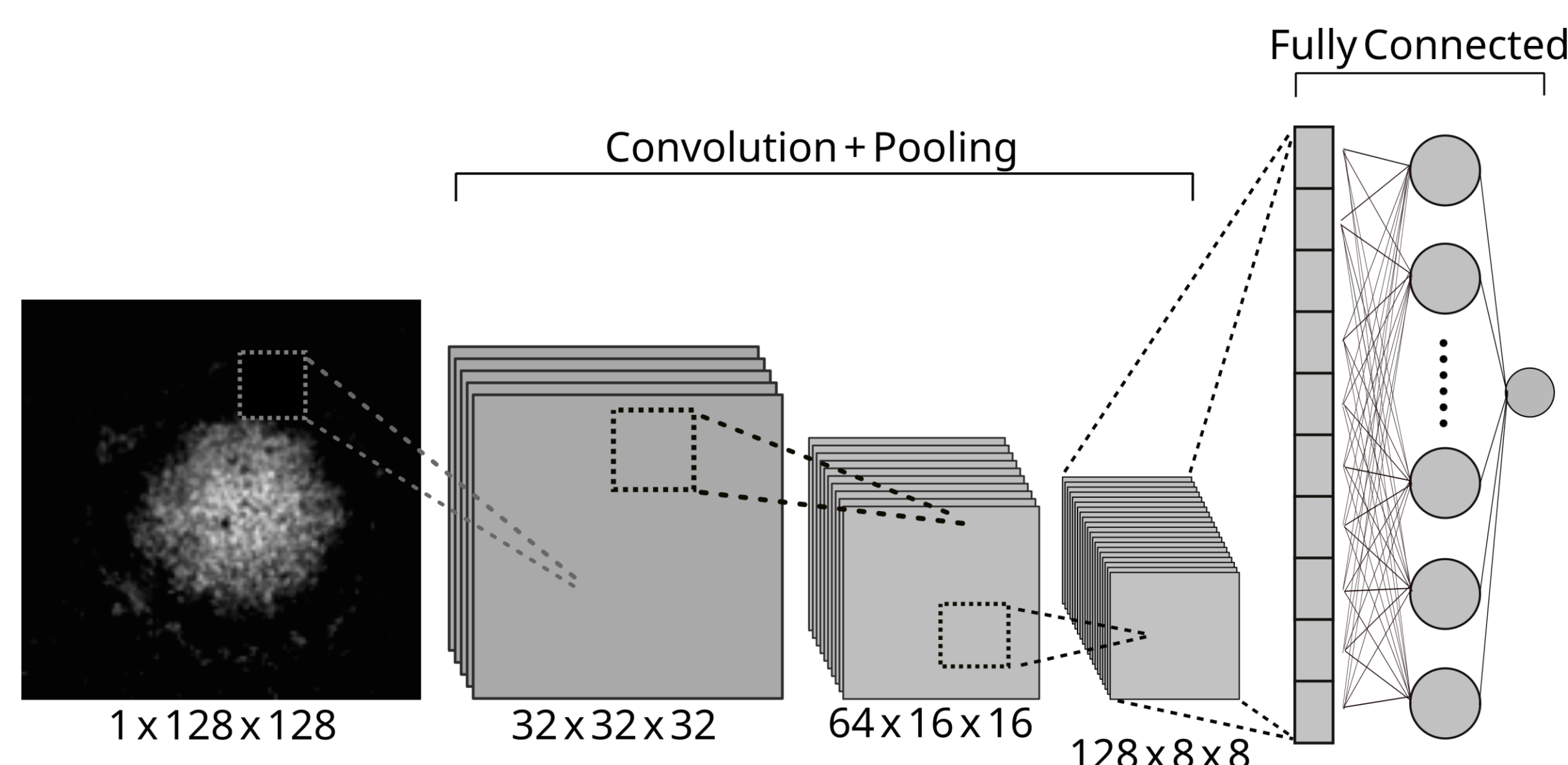


Figure 2. Schematic of convolutional neural network architecture.

① Euploid/Aneuploid Cells

② Healthy/Normal Lung

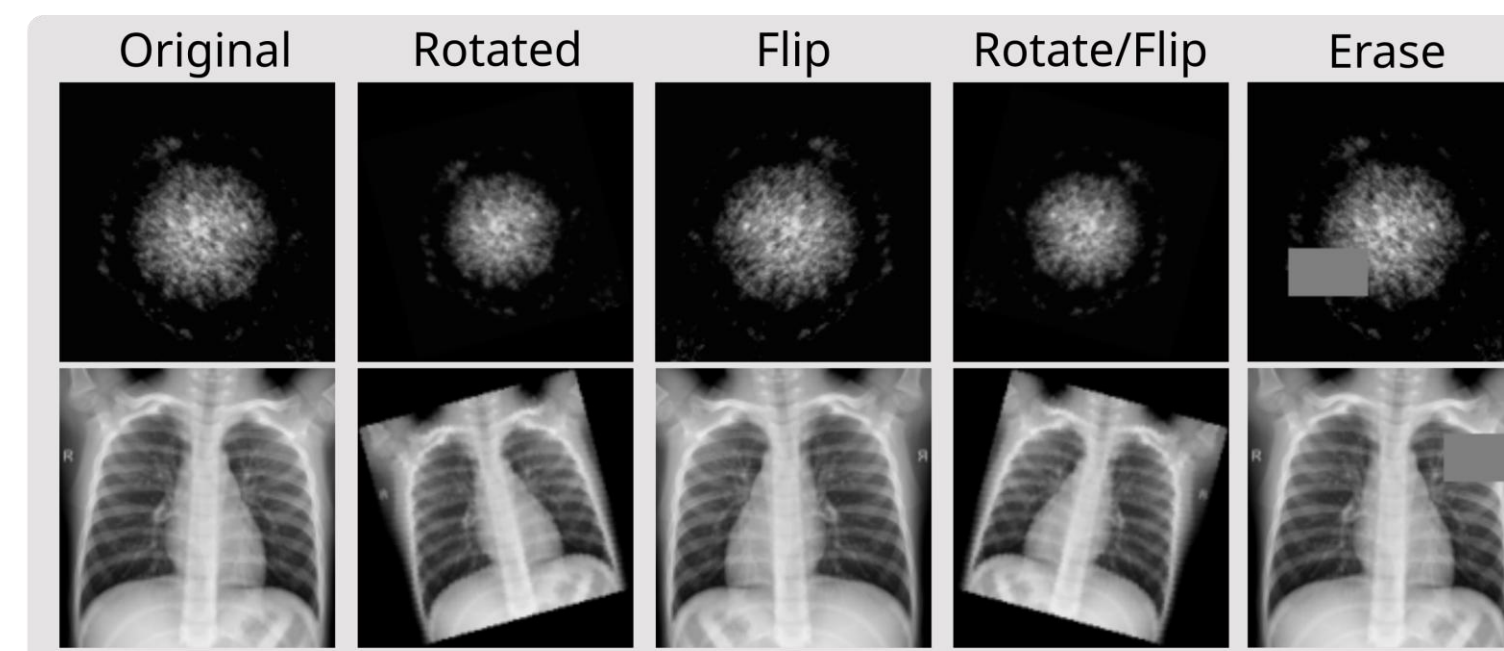


Figure 3. Examples of geometric transforms applied to training images

Results

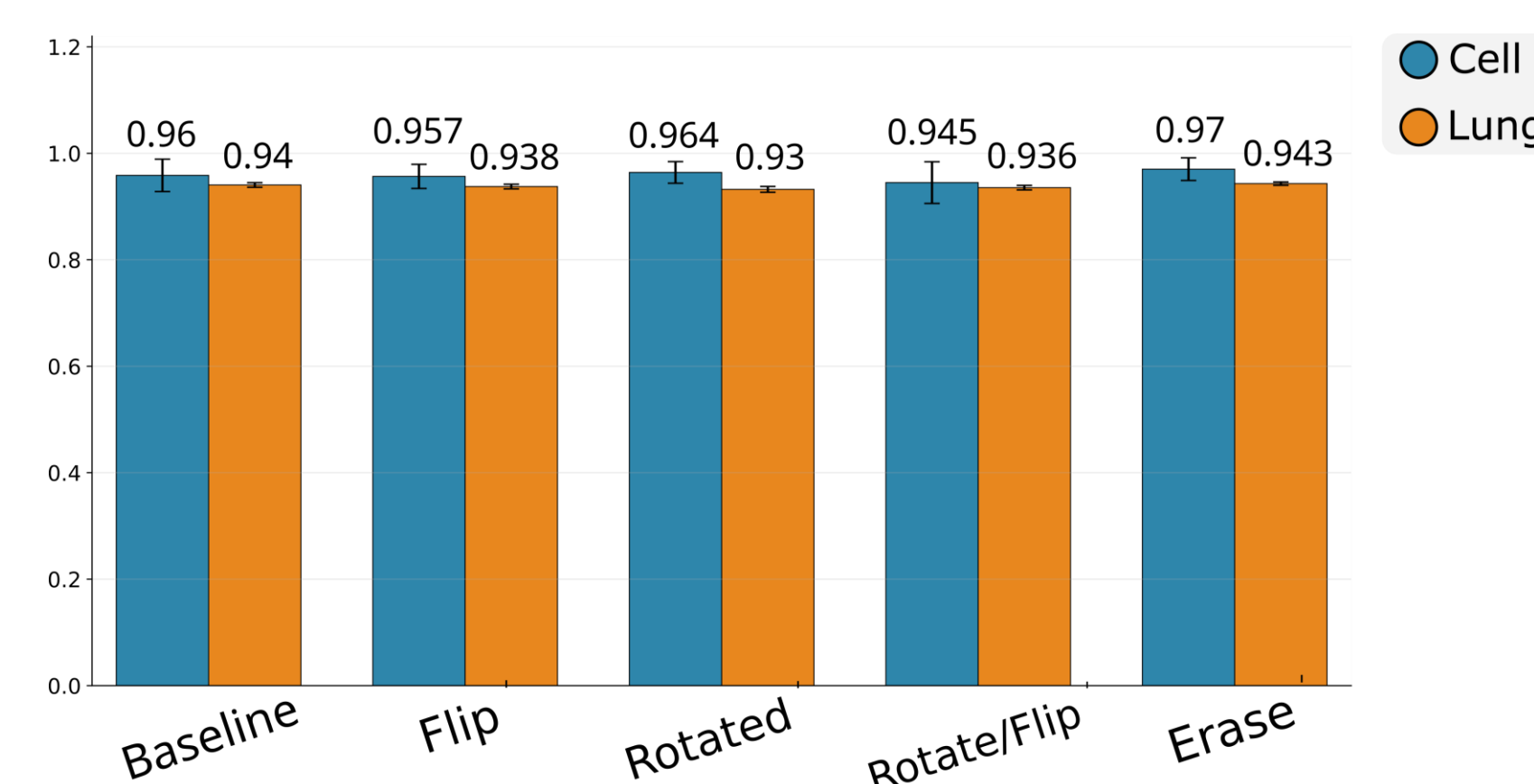


Figure 4. Mean test accuracy of model classification under different augmentation regimes. Bars denote the standard deviation across 10 seeds. Baseline experiment had no augmentations applied.

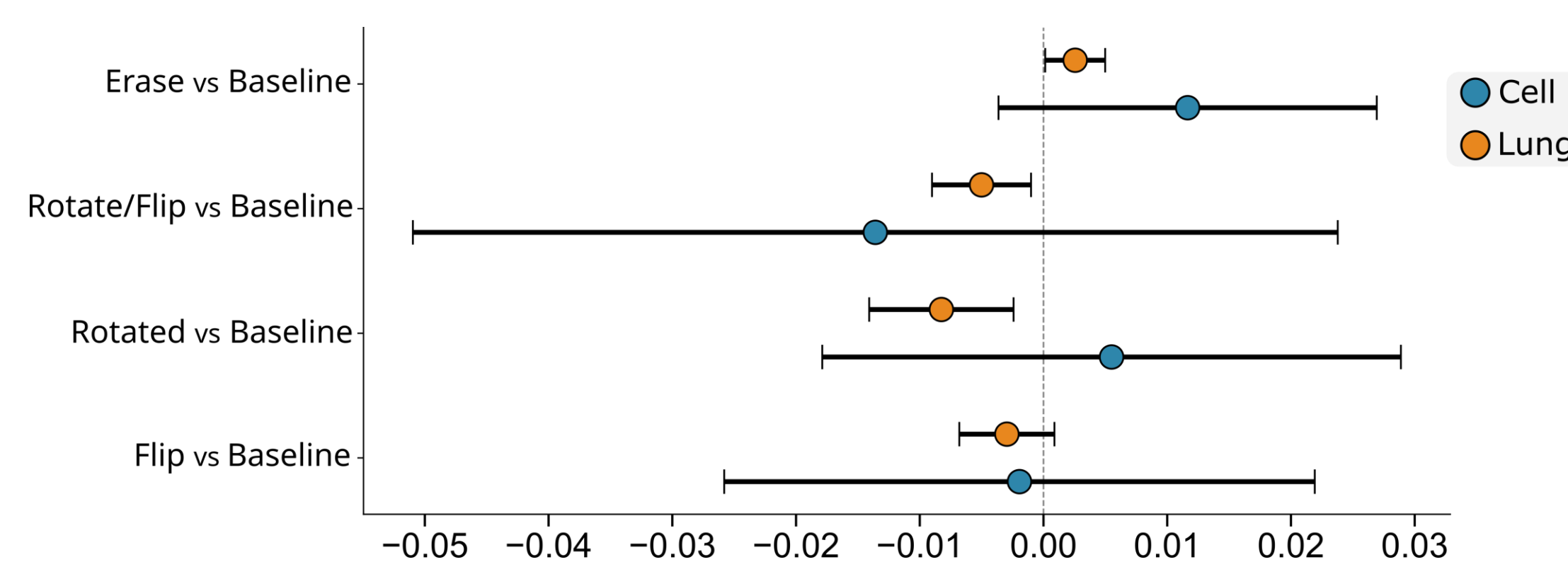


Figure 5. 95% Confidence Intervals for Δ Accuracy (Augmented - Baseline) across 10 seeds. Confidence intervals generated for both cell and lung dataset per augmentation.

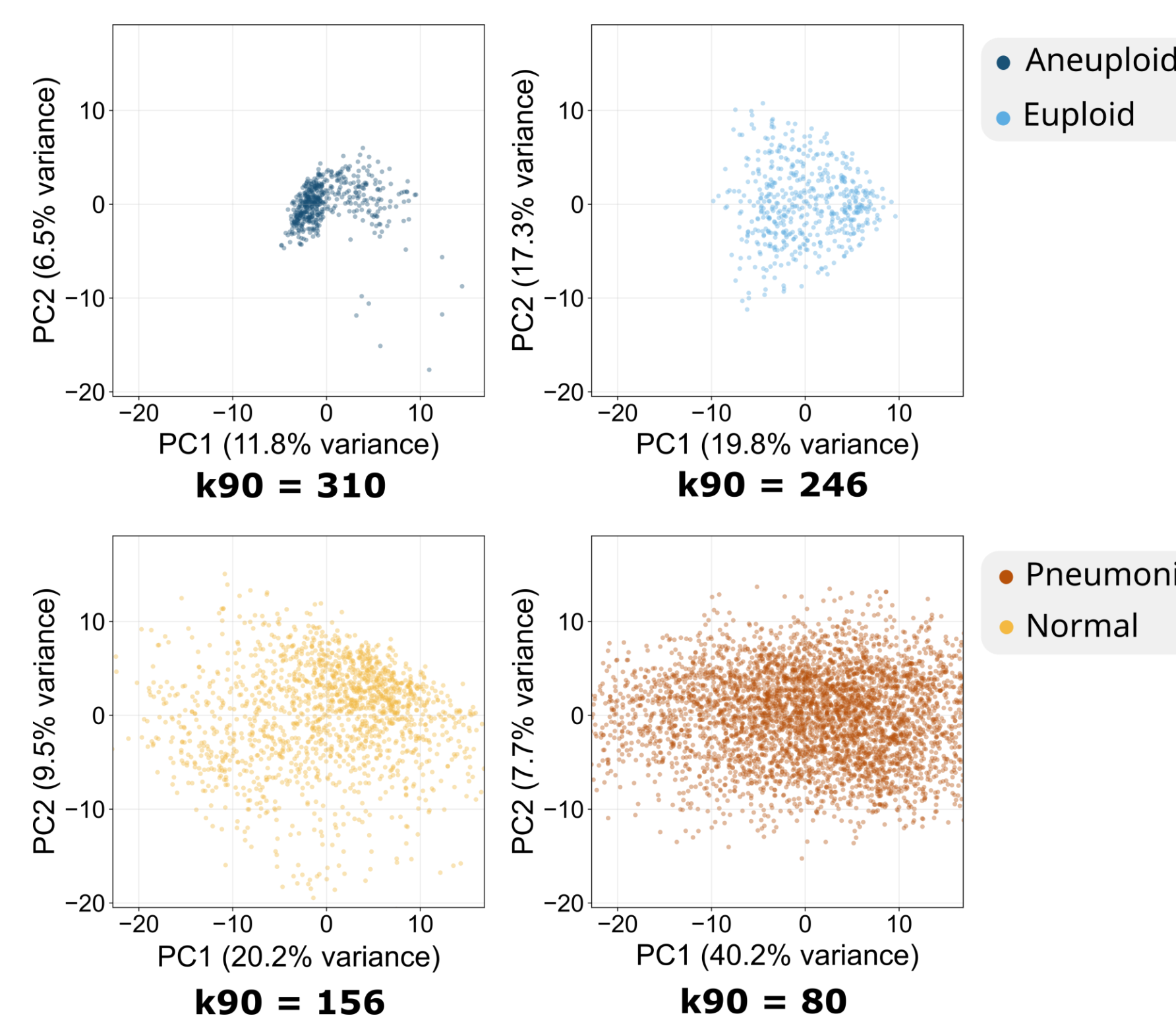


Figure 6. Intra-class 2d principal component analysis scatter plot. Each point represents one image projected onto the first two principal components. K90 values indicate the number of principal components required to explain 90% of intra-class variance.

Discussion

- The bar chart showed that all augmented data experiments performed comparably to the baseline. The overall cell accuracies ranged from 94.5% to 97%, with the baseline of 96%. The overall lung accuracies ranged from 93% to 94.3%, with the baseline of 94%.
- The accuracies showed that no augmentation experiment consistently exceeded the baseline performance for both cell and lung dataset, indicating no meaningful improvement from augmentation.**
- For cell dataset, all four augmentation experiments produced confidence intervals spanning zero, indicating no statistically reliable improvement over the baseline results.
- For lung dataset, the flip augmentation experiment also produced a confidence interval spanning zero. Rotation and Rotation with Flip augmentation produced confidence intervals entirely below zero, indicating a reliable decrease in accuracy relative to the baseline.
- Erasing on the lung dataset was the only augmented experiment that produced a confidence interval marginally above zero.
- Principal component analysis (PCA) revealed that the k90 ranged from 80 to 310 components, representing less than 2% of the possible pixel dimensions. That means images within each class occupy an extremely compact region of image space and exhibit low structural variability.
- The low intra-class variance helps explain the limited effectiveness of data augmentation. With training test distributions already containing little variability, the overall model may generalize enough without additional transformed data.**
- Under these conditions, augmentation techniques may introduce variation that does not exist in the test distribution, limiting the ability to improve generalization performance.

Conclusion

- Our results suggest that the underlying data distribution strongly influence the effectiveness of data augmentation techniques.
- These findings motivate more principled augmentation selection in biomedical machine learning pipelines.
- For supervised training, diversity analysis should be conducted prior to determine whether data contains sufficient structural variation for geometric transformations to provide genuine improvements.

References

- Goceri, E. (2023). Medical image data augmentation: techniques, comparisons and interpretations. *Artificial Intelligence Review*. <https://doi.org/10.1007/s10462-023-10453-z>